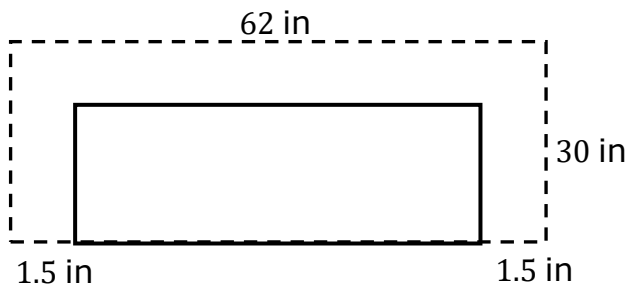


For questions 1-2, use the following scenario.

Vivian is putting in a new fence in her backyard. A scale drawing of her backyard is shown, with the dashed line representing where the fence will be installed. The scale used for the drawing is 2.5 inches = 8 feet.



1. How many linear feet is the fence around Vivian's house?

$$1.5 + 1.5 + 62 + 30 + 30 = 125 \text{ in} \qquad \frac{2.5 \text{ in}}{8 \text{ ft}} = \frac{125 \text{ in}}{x}$$

The fence is 400 ft.

2. The fence Vivian wants to install costs about \$18.50 per linear foot. How much will her new fence cost?

$$400 \cdot 18.50 = \$7400$$

3. A painter mixes paint to create a specific shade of purple. The ratio of each color in the mixture is 3 parts blue paint, 6 parts red paint, and 8 parts white paint. The painter makes a large batch of purple paint. He uses 10 gallons of white paint in the mixture. How many total gallons is the batch of paint he made?

$$\frac{6 \text{ parts red}}{8 \text{ parts white}} = \frac{7.5 \text{ gal red}}{10 \text{ gal white}}$$

Red: 7.5 gal + Blue: 3.75 gal + White: 10 gal

$$\frac{6 \text{ red}}{3 \text{ blue}} = \frac{7.5 \text{ gal red}}{3.75 \text{ gal blue}}$$

Total of 21.25 gallons

4. The largest deliverable pizza in the world is 54 inches wide, 54 inches long, and costs \$249.99 before tax. If 1 serving is 36 square inches, how much does one serving cost (before tax)?

$$54^2 = 2,916 \text{ in}^2 \text{ total pizza}$$

$$\frac{\$249.99}{81 \text{ servings}} = \$3.09 \text{ per serving}$$

$$\frac{2,916}{36 \text{ serving size}} = 81 \text{ total servings}$$

5. Kevin has determined that there are 136 calories in a $\frac{3}{4}$ cup serving of his favorite flavor of gelato. If Kevin eats $2\frac{1}{4}$ cups of gelato per week, how many calories is he consuming from gelato alone?

$$2\frac{1}{4} \div \frac{3}{4} = 3 \text{ servings per week}$$

$$136 \cdot 3 = 408 \text{ calories}$$

6. JoAnn averages 3,261 steps for 1.4 miles. She can walk 3.8 miles in an hour. How many steps can JoAnn walk in 90 minutes?

$$\frac{3.8 \text{ miles}}{1 \text{ hr}} = \frac{m}{1.5 \text{ hr}}$$

$$\frac{1.4 \text{ miles}}{3,261 \text{ steps}} = \frac{5.7 \text{ miles}}{s}$$

$$13,277 \text{ steps}$$

$$5.7 \text{ miles}$$

$$13,276.929$$

7. It takes Gal an average of 45 minutes to walk 2.3 miles to and from school each day. How many total hours will Gal have walked in 10 days?

$$2.3 \cdot 10 = 23 \text{ miles walked}$$

$$\frac{2.3}{0.75 \text{ hr}} = \frac{23}{h}$$

$$7.5 \text{ hours}$$

Ryan and Dwayne are making lemonade. Ryan's recipe calls for the 3 cups of lemon juice, 3 cups of sugar, and 9 cups of water. Dwayne's recipe calls for 6 cups of water, $1\frac{1}{2}$ cups of lemon juice, and $1\frac{1}{2}$ cups of sugar.

8. Will their recipes taste the same? Explain.

No because the proportion of lemon to water is not the same for their recipes.

Ryan

Dwayne

$$\frac{3c \text{ Lemon}}{9c \text{ Water}} \neq \frac{1.5c \text{ Lemon}}{6c \text{ Water}}$$

9. There are 16 cups in one gallon. How much lemon juice would Ryan need to make lemonade if he uses 1 gallon of water?

$$\frac{3c \text{ Lemon}}{9c \text{ Water}} = \frac{x}{16c \text{ Water}}$$

$$\text{Lemon} = 5\frac{1}{3} \text{ c}$$

10. How much sugar would Ryan need to add to 1 gallon of water to make lemonade?

Lemon and sugar amounts are in equal parts, so the sugar is also $5\frac{1}{3}$ c.